

Radar–Language Models (RLMs)

Foundations and Background

- Radar Sensing Fundamentals
- Physics-Informed Feature Extraction
- Related Fields and Traditional Radar Applications
- Large Language Models and Multimodal Extensions
- Integration Challenges

Categorization of RLMs

- Radar Signal Representations
- RLMs Architectures: Tokenization and Fusion
- Advanced Human-Computer Interaction (HCI)

Datasets, Assessment Techniques, and Implementation Methods

- Datasets and Data Synthesis
- Evaluation Methods and Reasoning Strategies
- Engineering Strategies
- Advanced Semantic Perception

Challenges

Signal sparsity & multimodal alignment

Data scarcity & annotation costs

Adversarial robustness & interpretability

Real-time deployment & latency constraints

Future Roadmap

The Era of Cognitive Radar Systems